Final Project Instructions

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CECS 361 - Digital Design Techniques and Verification

California State University, Long Beach

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Purpose

You are to perform an original project using your knowledge of HDL design & verification and utilize the Nexys A7 trainer board as your target technology.

This project will be conducted in groups of at most 3 members.

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Stage 1: Proposal

You are to create a project proposal that must include the following information:

- Title
- Group Members
- Abstract

Proposal is due on **wed, March 19th 2025**, 11:30PM on Canvas.

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Possible Project Ideas

Microprocessor/Microcontroller: Create your own simple processor (RISC, MIPS, etc.) and demonstrate the design by running a simple program.

Communication Protocol: Design and implement a communication protocol controller such as SPI, UART, etc.

Finite State Machine: Design an FSM or protocol to implement an advanced traffic intersection or an advanced configurable clock to show either military or standard time, VGA controller, etc.

Any architectural design that implements a solution to various hardware problems: Adders, multipliers, floating point ALUs, filters, etc.

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Useful Sources

"FPGA Prototyping by Verilog Examples: Xilinx Spartan-3 Version", by Pong P. Chu

"Fundamentals of Computer Architecture and Design", by A. Bindal

"Hardware Design Verification: Simulation and Formal Method-Based Approaches", by W. Lam

CSULB IEEE Xplore Database

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Stage 2: First Progress Report

The first progress report should include the following information:

- Introduction
- Background and Preliminaries
  - ❑ What research and resources are you using to assist you in completing your project?
  - ❑ Design diagram
  - ❑ What obstacles are you encountering?
- Implementation (20%-30% completed)
- Percentage and description of the work done by each group member

First report is due on **Friday March 31, 2024**5 11:30PM on Canvas.

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Stage 3: Second Progress Report

The second progress report should include the following information:

- Implementation (50%-60% completed)
- Progress
  - What changes have you made to your project since the first progress report?
  - What challenges are you encountering?
- Percentage and description of the work done by each group member

Second report is due on **Wed, April. 14, 2025**, 11:30PM on Canvas.

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Stage 4: Final Report

The final report should include the following information:

- Title, Group Members, Abstract, Introduction, Background and Preliminaries, Implementation (100% completed)
- Evaluation
  - ❑ Were you able to complete your project as intended/results?
  - ❑ What steps/techniques did you take to verify your project? (Include the simulation waveform for different modules and the snapshot of the running board.)
- Conclusion
- References

There is no page requirement, but the report should be comprehensive.

Final Report should be formatted following IEEE Format guidelines: https://www.ieee.org/conferences/publishing/templates.html

Final report is due on **Monday, April. 30th, 2025**, 11:30PM on Canvas.

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Stage 5: Presentation

Present your project and demonstrate your results (8-12 minutes per group followed by 3 minutes Q&A.)

Presentations are to be conducted on the following dates:

May 6th and May 8th

Satisfactory completion of the final project (obtaining at least half of the final project points) is a requirement to pass this course.

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Grading

The projects will be graded based on the following rubric:

- Presentation (25pt)
- Report Quality (25pt)
- Project Quality (25pt)
- Implementation (25pt)

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